FEATURE ARTICLE

# Five Lessons for Applying Machine Learning

*To make the most of machine learning, organizations will need to take note of five key lessons.*

**Robbie Allen**

Machine learning is a type of artificial intelligence; in machine learning, computers learn patterns from data. It is a transformative technology for the business world. But currently, businesses are caught in a net of misunderstandings, confusion, and knowledge gaps about machine learning. As the CEO of machine learning company Infinia ML, I want to help close these gaps.

Our team uses machine learning to automate data-driven business processes. In other words, we use it to make real business impact. While many on our team have impressive academic credentials, we don't do science experiments for their own sake. We're focused on the real outcomes our project can deliver; in fact, we're writing a book called *Machine Learning in Practice*. Here, I offer five critical lessons we've learned that can help elevate the conversation around machine learning and accelerate its adoption in the enterprise.

## Lesson 1: Machine Learning Is Not Overhyped

There's no question that machine learning is the subject of a lot of hype. In fact, machine learning or a variant term has been at the peak of Gartner's Hype Cycle for four years in a row. But I don't expect it to fizzle out. Machine learning might take a long time to mature, and it certainly won't meet all of people's expectations. But it is here to stay, for at least two reasons.

The first big reason for the power of machine learning is the development of deep neural networks and deep learning. Although machine learning has been around for a while—the term was coined in 1959—it's taken a long time to reach today's media ubiquity. In part, that's because newer techniques can take advantage of the power of graphical processor units (GPUs). The rise of GPUs is due in large part to the game console era. Machines like the Xbox and the PlayStation, and the graphics-intensive games they run, needed high-powered GPUs, so chip makers started making more of them. We can do more with machine learning today because we have access to the power of these processors.

The other factor is data. The big data era had to precede the machine learning era, because machine learning runs on data. You have to have data—typically a lot of data—to do interesting things with machine learning. That's especially true for deep learning, a more complex version of machine learning. Deep learning models tend to be data hungry. Fundamentally, machine learning and deep learning systems find patterns in data. If you're developing a machine learning solution, you need to provide the algorithm with a training dataset that is representative of the patterns you want to be able to find when it goes to production.

These advances in techniques, processing power, and data access are driving a lot of new work and interesting progress in image processing (specifically object recognition), natural language processing (NLP), and complex pattern recognition (see text box "Advances in Machine Learning Applications," p. 40).

I believe we could stop innovating in machine learning today—no new research papers, no new techniques—and there would still be a 10-year backlog of machine learning technologies and approaches to deploy in most companies. The challenge is that most companies don't have the technical expertise and understanding to take advantage of machine learning. That will change over time, as the needed skillsets become more prevalent. We're already starting to see improvement there. As the skill sets develop, the applications will become more diverse.

We are already seeing some interesting and valuable applications in a range of fields. One example is in the automated contract analysis space. A leading legal services firm came to Infinia ML looking for technology to automatically sort the documents by type—is it a master service agreement? a statement of work? an amendment? a nondisclosure agreement?—and identify potential issues with change of control, such as indemnification clauses. Maybe that sounds like an easy thing, but when you're trying to create a system to apply these criteria generically, you find that there are lots of

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As CEO of Infinia ML, Robbie Allen helps companies understand and implement machine learning.

different ways to talk about change of control. A simple search for "change of control" probably won't do it.

This is a problem that machine learning actually works well for, because you can train an algorithm to recognize the many different ways of specifying change of control. The algorithm can learn those patterns and detect them even when they're not specified in the exact same way each time. Machine learning is going to help the firm transform its business from a predominantly people-focused business to a combination of people and technology.

Another example is an ultrasound device for mobile phones. This kind of application has all sorts of interesting implications for healthcare accessibility; it could be a global game changer. The device is made by a Chinese company; it plugs into an Android phone. It's actually pretty hard to find things on an ultrasound—a particular artery or organ, for instance—especially without the highest-resolution technology. Ultrasound technicians get a lot of training in reading these images. Infinia ML developed a mobile version of deep learning that could detect objects in an ultrasound image.

Infinia ML has also worked with a company called Align Technology, which makes Invisalign aligners. The big challenge for that market is the high barrier to use the product. To really understand the impact of Invisalign, you have to understand its process, which requires going to an orthodontist and having a mold made of your mouth. It's a bit of an invasive process just to understand whether the product would work for you. What Align wanted was a mobile phone app that could provide some insight into the value of the product instantly. The idea was that the customer would take a selfie and the app would use that picture to show what Invisalign would do after treatment.

Again, that's a machine learning problem. It's a vision problem. And it's a bit harder than you might think. The app has to find the teeth in the picture; it has to represent the likely change fairly accurately, or the company could get in trouble with consumers. In fact, it has to be able to find where each specific tooth is and model how its position could change. Underlying this simple app that gives you an instant "after" shot is a really complicated machine learning application.

These examples illustrate a key point: machine learning is often not going to be implemented in a way that smacks you over the head. In the next 5 to 10 years, machine learning will fade into the background; it'll be a technology like databases are a technology. You won't hear about companies, like mine, that are leading with machine learning because it'll become much more commonplace. Ultimately, it will be a fairly routine transformation for most people. Machine learning will have significant impact, usually without the consumer or the customer even knowing it.

**Lesson 2: Machine Learning Is Still Immature**

This might seem to contradict the first lesson, but although the hype may be off the charts, the actual maturity of the underlying technology is relatively low. The tools and platforms that are available today to get machine learning into production are still relatively immature.

Everyone thinks they need to get on the machine learning bandwagon because everybody else is doing it. That's not exactly the case. There are a lot of people trying it and dabbling with it, but actual real-world production implementations are just starting to be available. If you're just starting your machine learning journey, now is a good time. You're actually not that far behind. That won't be the case in three or four years. Things are going to accelerate quickly, and we're starting to see lots of companies go down that path.

**Lesson 3: Your Data Are Not Ready for Machine Learning**

Having worked now at two data-driven artificial intelligence companies, I've noticed a common phenomenon: clients' data

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## Advances in Machine Learning Applications

*Image Recognition*

When you go to the airport, you put your bags on the scanner and somebody from TSA looks at a screen to try to determine whether you have a knife, gun, or explosive in your bag. That system is suboptimal. One alternative is to use deep learning to detect weapons in baggage. Infinia ML's Chief Scientist, Larry Carin, has demonstrated a system that analyzes images of every bag to look for threats. When it identifies something suspicious, it generates a notification to prompt a human being to take a look. This approach to object recognition is useful in a whole host of areas, and the technology, especially the machine learning component, has gotten really good.

*Natural Language Processing*

Traditionally, NLP has been a brute-force technology that has not worked well, because languages are not very specified: it's very difficult to articulate all the semantics of a given language, since to do so you would need to generate a comprehensive model. If you multiply that across all the different languages, each with its own semantics, you can see how difficult the problem is. Many companies that tried to advance NLP prior to machine learning didn't have great success.

With the introduction of machine learning in the NLP space, we're seeing significant advances in capabilities such as machine translation. The brute-force approach to machine translation, the phrase-based method Google had been working on for many years, had gotten pretty good. But adding the neural approach brought a step-wise increase in performance; at least for some language pairs, machine translation is approaching human-level performance. It will only improve from here. It's likely that we'll have human-level performance across most language pairs in the future.

There are other applications in the NLP space as well, like chatbots. These have also improved markedly over just a few years, and it is all due to machine learning.

*Complex Pattern Identification*

Machine learning can help identify patterns in many different types of data—numerical data, unstructured data, even text data. Take, for instance, Google's AlphaGo project, which is software that beat the best grandmasters at the ancient Chinese game of Go. That system takes in lots of data and uses it to predict the pattern of moves that will win in a particular game. Or take systems that try to identify optimal applicants for a particular job. Identifying the likely best performers from a pool of applicants is a complex pattern identification problem.

isn't ready, and often they don't know it. I can't really blame the clients, because they don't really know what it means for data to be ready for machine learning.

As much as a third of the overall life cycle of a project is spent on getting the data cleaned up—maybe even more in some cases, depending on the datasets. Why don't companies know that the data needs cleaning when professionals in machine learning can see it very quickly?

Consider what it's like to get your yearly physical. You would think that you know your body better than anybody, so what is a doctor going to be able to tell you? Well, the doctor performs a variety of tests, checks some things you probably don't do on a regular basis, and from that may come back with a diagnosis that you didn't expect.

It's a very similar process when data scientists look at a dataset. You might think you know your data better than anybody, but you're probably not poking and prodding at it in the same way a data scientist does in creating a machine learning solution. You don't know what you don't know. You don't know what we're looking for.

At Infinia ML, we evaluate data for a machine learning project based on a number of factors, including:

- Is the data accessible?
- Is the data sizeable enough?
- Is the data useable?
- Is the data understandable?
- Is the data maintainable?

***Is the Data Accessible?***

When people think of problems that they want to solve, they may know the data they need exists somewhere—they may even know where it exists—but sometimes, they can't actually get to it. That's almost like it doesn't exist.

The data may be siloed in another part of a large bureaucracy. Or it may be technically accessible but legally off limits. Think about customer data. Often companies come to us wanting to do some interesting analysis on customer data. Our first question is, do your terms of service allow for that usage? If the terms of service were created 10 years ago, before this kind of data usage was contemplated, it may be a gray area. The issue is whether your customers are okay with you using the data in the way you plan to use it. In some cases, you may have to go back and ask for permission from clients or change your terms of service. It's not always a given that just because you know where data exists, you can actually use it.

***Is the Data Sizeable Enough?***

Machine learning is data hungry. You need a fair amount of data to do machine learning.

How much is enough? Unfortunately, there's no easy answer for that question—it depends on the specific problem you're trying to solve. Typically, it's not in the hundreds of samples; it's in the thousands at least, and tens of thousands is much better. If you're getting into millions of samples, that's likely even better.

Often the amount of data you need depends on the patterns you're trying to identify. If you're trying to do a binary classification—it's either in this bucket or that bucket—you probably won't need as many samples. If you're trying to make a prediction across 10 different buckets, and the distribution is fairly similar across those, then you need a lot more samples. The requirements will vary.

How much data you need also depends on whether you are able to apply data from another domain to your problem. The TSA project I mentioned sometimes is a good example of this. The initial algorithms for the TSA X-ray deep learning system were based on some well-known natural image object detection algorithms that were built on large corpuses of natural images—pictures of cats and dogs, for instance. Google has access to lots of these types of images. There are also well-known datasets with millions of natural images. It turns out you can build a model from those images, apply it to a much smaller subset of images for a specific domain like X-rays, and the performance of the algorithm doesn't suffer. That's a place where you can use a base dataset that actually isn't your dataset, apply a layer of your data on top of it, and get the same performance as if you'd built the entire model on your data.

***Is the Data Usable?***

One of the top problems we face as data scientists is a dataset with hidden problems. There might be holes in the data, for instance. The customer tells us that a column should be fully populated, and it's not. But because we assumed that it was populated, something breaks. That's very common. Or there might not be a common understanding of the format or syntax of the data in that column.

***Is the Data Understandable?***

A data scientist will get a spreadsheet that may have 50,000 rows in it, but the column headers are "Column 1," "Column 2," "Column 3," and "Column 4." Those headers don't provide the information we need to make sense of the dataset, determine what the boundary cases are, and understand what we can expect the dataset to include. This kind of information is often lacking. Even if you've been doing a pretty good job collecting data, and you have a big, complete collection of data, the data might not be available in a form that communicates the contents of the data file to a third party or even to internal people.

***Is the Data Maintainable?***

Can you replicate the process of producing the data? For instance, say your company is working with Infinia ML. We'll ask for a dataset that has specific attributes, and you'll cull together the data from a couple of different places, do some transformations on the data, and provide us a CSV file with a million rows in it. It might have taken you a couple of weeks to put that data together. We do our work on it, we do some training, we come back with some results, and we're ready to put the system into production by feeding it live data. Now you have to think through how you got that dataset to begin with. Can you repeat that process consistently over time? You'd be surprised how many times we hear, "Well, actually, I'm not sure we can do that same thing in production. We have the data, but that's just because I hacked it together."

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From the beginning, when you're trying to create datasets or doing transformations on the data—even just for a test run—make sure you're doing it in a way that is repeatable and maintainable. If you can't create a maintainable dataset, it's as if the data doesn't exist; the system won't be usable beyond the training mode.

### Lesson 4: Centralize Your Data Science Resources

This is something that I hadn't really thought about before starting Infinia ML, but it's become obvious in hindsight.

Software engineering is typically distributed across a company. You'll have engineers in different departments, and you don't have to necessarily centralize those resources. Data science is often distributed in the same way. Each department may have its own data scientist, or even a couple of data scientists.

That presents two problems. One is that data scientists like working with other data scientists. At Infinia ML, we've received over 700 applications for our data science position. I ask everybody we bring in, "Why were you interested in Infinia ML? What is it that drew you to the company?" And they all say, "I went to your web page, and I saw that you have all of these data scientists; I thought I could learn a lot from that team." Often, they are the only person at their company doing data science, and it's hard to be the person that has all the answers. If you're looking to hire one data scientist, it may be difficult to find that person.

Part of the reason for this is related to the second challenge: data science is still a really immature field. It's not like software development, where a software developer at one company mostly resembles a software developer at another. People are still trying to figure out data science. That means practices vary across companies. Data scientists recognize that, and so any opportunity they have to work with other data scientists is attractive.

It's not necessarily about having lots of data scientists. Many companies have more data scientists in total than Infinia ML does—we have 16 at the time of this writing. But there are very few companies that have a single team of data scientists larger than ours. Centralizing your data science function can allow your data scientists to learn from one another and possibly help you attract more good data

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scientists. Centralization may not be necessary forever, but while we're still figuring out what it means to be a data scientist and developing data science practices, it's absolutely critical.

### Lesson 5: It's Not a Jobs Problem—It's a Skills Problem

I do a lot of interviews to discuss the workforce challenges AI is presenting and the general worry that AI is going to steal everybody's jobs. I'm here to say that AI will not create a jobs problem, but it will create a skills problem. At least for the foreseeable future, AI and machine learning will not disrupt the workforce any more than we're used to with new technologies. To be sure there is always disruption happening in the workforce. There are always businesses in decline that result in people losing jobs. That will continue to happen, and machine learning may be a small part of that. But a wide-scale loss of jobs and wide-scale job shortage is not going to happen anytime soon.

Let me give you an example. My last company, Automated Insights, had a platform that created written content at a large scale. A few years ago, we produced more than a billion and a half pieces of content in a year. Automated Insights powers the Yahoo and NFL fantasy football recaps, for example—80 million of those every year. Journalists love to call me up and talk about the future of journalism, because they saw this and thought, "Wow, you can produce something that a journalist typically provided, so does that mean that you're going to automate journalists out of a job?"

But despite all of those interviews over the years, the actual number of jobs lost due to an implementation of our platform is zero. This is true across applications, including the creation of automated quarterly earnings stories that Associated Press reporters had written manually in the past. Nobody lost a job; in fact, journalists were glad not to have to do that task. Likewise, in the 20 machine learning projects we have done over the past year at Infinia ML, no one has lost a job due to implementation of a machine learning system.

Why is this? Take the TSA example. Do we really think that having a person stand behind a monitor and look for tiny things on an X-ray screen is the best use of a human's abilities? If we had technology that could do that, it would be significantly better than having a person do it. A lot of the first jobs that are going to be automated are going to be these kinds of jobs—work that should never have been done by a person to begin with.

Further, that a task is automated does not mean there is no further use for the person who used to do that task. For instance, if baggage scanning ever becomes 100 percent human-free, those agents could be freed up to watch for behavioral cues or to help travelers get through the line faster.

In fact, machine learning and AI will create a lot of jobs. We're seeing in the machine learning life cycle a whole new set of jobs, including what I call machine learning assistants—people who continually train the algorithm. It's not really machine learning if you stop training the algorithm once it's in production; you need a person to guide the algorithm over time, providing a feedback loop so the algorithm can continue to learn. And that's just one of a whole set of new types of jobs that will be coming in the future.

Machine learning won't just come in and take jobs; in fact, it may move some jobs up the value stack, for instance, from roles strictly focused on data entry to roles that help improve the accuracy of the algorithm's solution. In some ways, there may be a bright future for jobs, a future in which people provide more value because they can participate in providing the solution rather than just entering data or doing other repetitive tasks.

The problem is that those jobs will require different skills. I said earlier that there's a 10-year backlog of machine learning developments to implement. The primary reason for this backlog is that we don't have enough people who understand machine learning. That is a big challenge.

You'll start to see more solutions to the skills gap problem in the next few years, but frankly, there are not a lot of great solutions right now. There are all sorts of online materials, through organizations like Coursera and Udacity; there are many training classes you can take online. But that's not exactly what's needed. What's lacking is opportunities for experiential training—opportunities to actually try out the science and figure out how it works on the ground. You need that real-world apprenticeship, like a doctor's residency, to be an effective data scientist.

### Conclusion

Machine learning will bring major changes to the business world in the coming years. Here are three of my predictions about the future:

1. Jobs that require manual visual inspection will start to go away within a few years. The technology's getting so good now that we won't want humans' assessments based on their inspections of an image or video. That kind of work will largely be replaced by machine learning technologies.
2. In a few years, all document analysis jobs will also have a heavy machine learning component, as advances in NLP make those capabilities more reliable and more powerful. Again, examining large sets of documents and trying to answer questions about those documents is not something people are ideally suited to do. But that's exactly what machine learning–based NLP technology is good at doing.

3. The big challenge will lie in extracting the data from existing systems and finding ways to manipulate it. Infinia ML's executive chairman, Mike Salvino, was the former Group Chief Executive of Accenture Operations. He sees this challenge as the key to the future. Companies will have to spend as much money getting data out of systems like SAP, Oracle, Cerner, Epic, or Salesforce as they spent installing them in the first place. Those data collection systems were not really built for machine learning integration; they were not built to make it easy to get the data out and manipulate it in interesting ways. The next generation of technologies that come behind the current systems will be much better suited to handle that type of task.

To these three predictions, I'll add one more that I feel is a near certainty: the companies that are able to put machine learning into practice effectively will dominate the future. Those will be the companies that have already started their machine learning journeys or that make a decision to start now. These companies won't just be data driven; they will master that data with machine learning.

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